



DATA SCIENCE CAPSTONE PROJECT

Diabetes Prediction

By Matthew Madriaga, Ray Gomez, Brayden Davies, Seavchhun Teng





Overview

Objective

Predicting HbA1c
and Diagnosing
Diabetes
Validity of ANNs

Background

What is HbA1c
and Diabetes?

Statistic

New Zealand (2024)
348,000
Diabetic cases





Methodology



No Missing Data



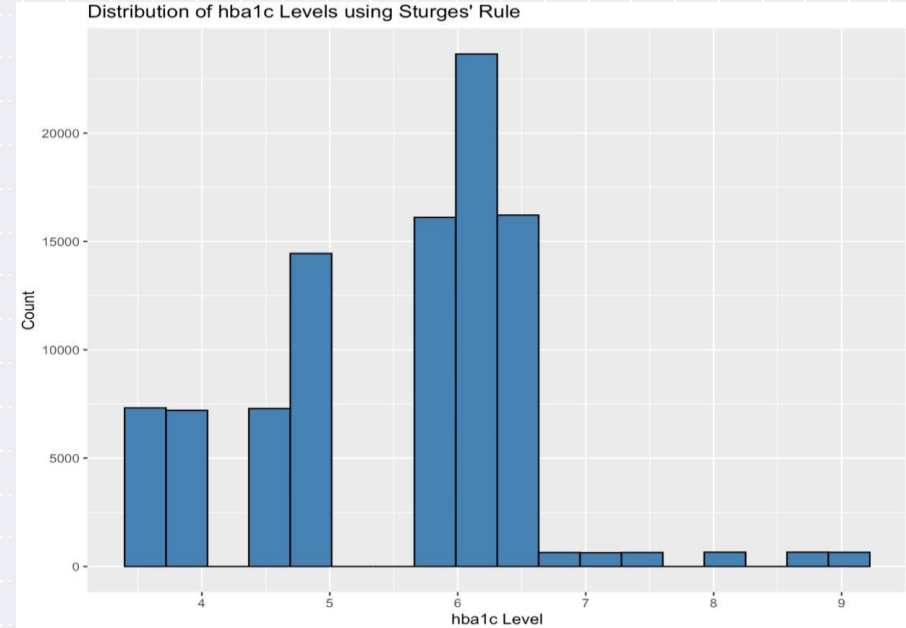
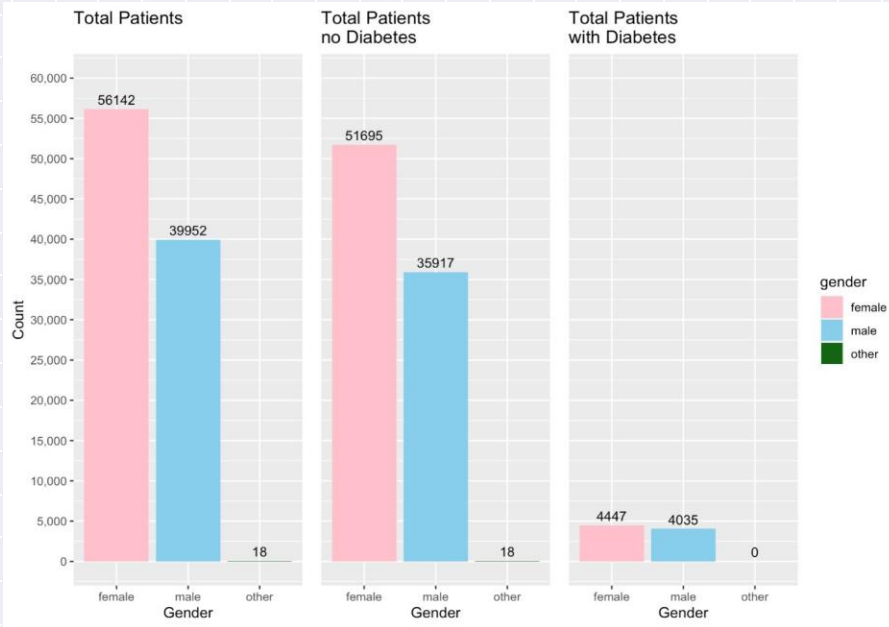
Class Imbalance Categorical Data



Approximately **8%** Duplicates



Right-Skewed Numerical Variable





HbA1c Levels Prediction

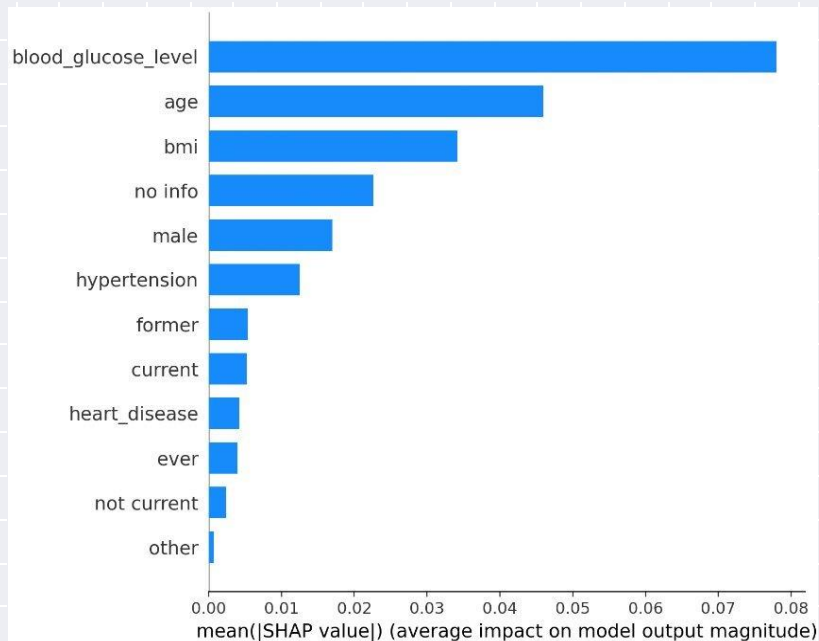
ANN Model		RMSE	MAE	R2	Imp vs Mean	Imp vs Median
Original Model	age + bmi + blood glucose level	1.042	0.876	0.064	6.4	11.8
Adding Features	smoking history	1.043	0.878	0.061	6.1	11.6
	smoking history + heart disease	1.043	0.873	0.062	6.2	11.7
	smoking history + heart disease + hypertension	1.042	0.876	0.063	6.3	11.7
	smoking history + heart disease + hypertension + gender	1.042	0.874	0.063	6.3	11.8
	heart disease	1.040	0.876	0.066	6.6	12.1
Best Model	heart disease + hypertension	1.041	0.875	0.065	6.5	12
	heart disease + gender	1.041	0.876	0.065	6.5	12
	heart disease + hypertension + gender	1.041	0.873	0.065	6.5	11.9
	hypertension	1.040	0.873	0.066	6.6	12.1
	hypertension + gender	1.042	0.873	0.063	6.3	11.8
	hypertension + smoking history	1.043	0.88	0.061	6.1	11.6
Feature Interaction	age * bmi * blood glucose level	1.043	0.876	0.062	6.2	11.7

Chosen Model, RMSE = 1.041, MAE = 0.875, R-Squared = 6.5%

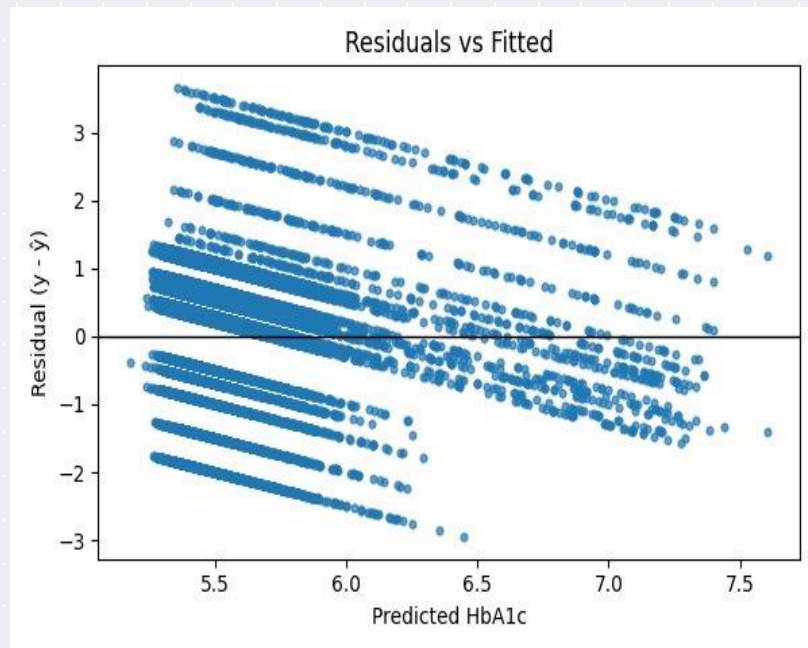


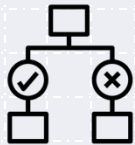
HbA1c Levels Prediction (Continued)

SHAP Feature Importance (ANN)



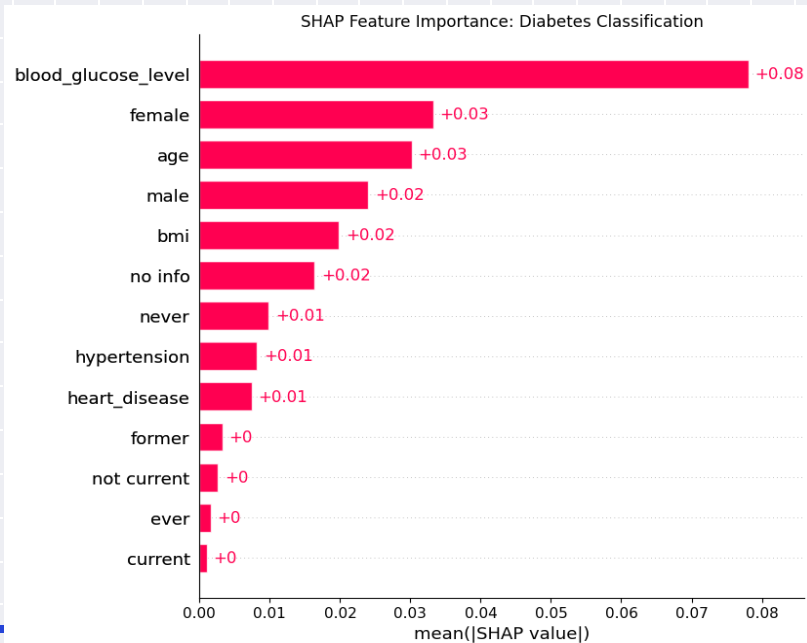
Optimal Model's Performance



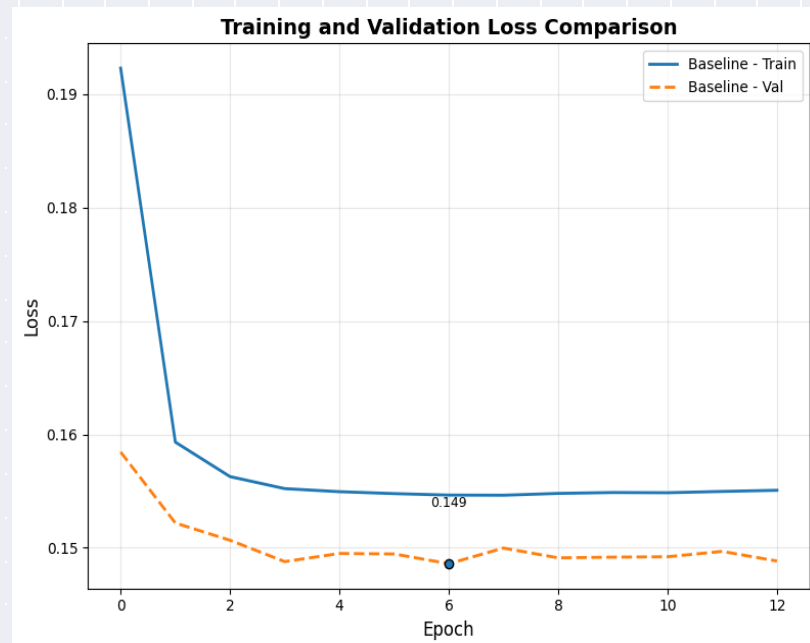


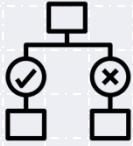
Diabetes Classification

SHAP Feature Importance (ANN)



Optimal Model's Performance





Diabetes Classification (Continued)

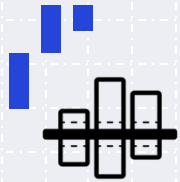
ANN and Logistic Regression were evaluated using

Accuracy, Specificity, Sensitivity, and AUC

? — What is Specificity and Sensitivity?

Classification Table

	ANN	Logistic
Accuracy	0.83	0.83
Specificity	0.83	0.84
Sensitivity	0.84	0.78
AUC	0.92	0.91



Gender Differences

Gender-Based analysis used same predictors and threshold as optimal ANN model

Classification Table

	ANN		Logistic	
	Male	Female	Male	Female
Accuracy	0.79	0.84	0.81	0.85
Specificity	0.78	0.84	0.81	0.86
Sensitivity	0.86	0.85	0.80	0.77
AUC	0.92	0.93	0.90	0.91



Male Model: Better Sensitivity



Female Model: Better Specificity





Real World Application



HbA1c Prediction ANN Model's Key Usage

Early Flagging, Preventative Measure



Diabetes ANN Model's Features of High Importance

Blood Glucose, BMI, and Age



Conclusion



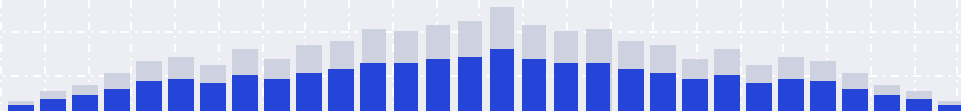
HbA1c Prediction: Weak Prediction Model



Diabetes Prediction: ANN performed better than logistic regression



Gender Based Analysis: Female performed better





Thank you
Any Questions?
